INCH-POUND
MIL-DTL-24643/81
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, TOPSIDE, MULTI-CONDUCTOR, TYPES LSE1E, LSE2E, LSE3E, LSE4E, LSE7E, LSE10E, LSE14E, LSE19E, LSE24E, LSE30E, LSE33E, LSE37E, LSE44E, AND LSE61E
(INCLUDING VARIANTS LSE1EW, LSE2EW, LSE3EW, LSE4EW, LSE7EW, LSE10EW, LSE14EW, LSE19EW, LSE24EW, LSE30EW, LSE33EW, LSE37EW, LSE44EW, AND LSE61EW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSE**E (non-watertight) and LSE**EW (watertight).

Construction (Non-Watertight and Watertight)

First - Component wire in accordance with MIL-DTL-24643/80 (see table I or table II for size).

Second - The required number of conductors (see table I or table II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded cable assembly (single-conductor cables are exempt from the cabling requirements stated above). Conductor identification codes shall be applied using method 1 or 3. If method 1 is used, outer surface to be printed shall appear black. Single-conductor cables shall be colored white or natural with no other identification applied.

Third - An optional binder.

Fourth - Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).

Fifth - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

Sixth - Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape shall provide electrical contact between the inner and outer braided shields.

Seventh - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG. For watertight constructions only, the use of solid or semi-solid water-block materials (such as silicone, hot melts, or gels) is not allowed on the outer braid. Watertightness shall be by means that could include, but are not limited to, water-swell yarns incorporated within or over the braid or water-swell tape helically applied over the outer braid with 25 percent minimum overlap. The water-block material shall not stick or adhere to the outer braid strands.

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Eighth - An opaque optional binder (tape shall not be clear).

Ninth - Cross-linked polyolefin jacket. See table I or table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Details (non-watertight).

Military part no. M24643/81	Type and size	Conductor size (AWG/ MCM)	Number of conductors	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)
					min.	max.	
-01UO	LSE7E-2	18	7	0.045	0.480	0.530	7.070
-02UO	LSE10E-2	18	10	0.045	0.585	0.650	7.070
-03UO	LSE14E-2	18	14	0.050	0.640	0.705	7.070
-04UO	LSE19E-2	18	19	0.055	0.715	0.780	7.070
-05UO	LSE24E-2	18	24	0.065	0.845	0.915	7.070
-06UO	LSE30E-2	18	30	0.070	0.895	0.970	7.070
-07UO	LSE37E-2	18	37	0.075	0.970	1.055	7.070
-08UO	LSE44E-2	18	44	0.085	1.100	1.185	7.070
-09UO	LSE2E-3	16	2	0.045	0.390	0.440	4.400
-10UO	LSE3E-3	16	3	0.045	0.410	0.460	4.400
-11UO	LSE4E-3	16	4	0.045	0.440	0.490	4.400
-12UO	LSE7E-3	16	7	0.045	0.510	0.560	4.400
-13UO	LSE10E-3	16	10	0.050	0.635	0.700	4.400
-14UO	LSE14E-3	16	14	0.050	0.690	0.755	4.400
-15UO	LSE19E-3	16	19	0.060	0.770	0.845	4.400
-16UO	LSE24E-3	16	24	0.070	0.915	0.990	4.400
-17UO	LSE30E-3	16	30	0.075	0.970	1.050	4.400
-18UO	LSE33E-3	16	33	0.080	1.010	1.095	4.400
-19UO	LSE37E-3	16	37	0.090	1.055	1.135	4.400
-20UO	LSE44E-3	16	44	0.090	1.195	1.280	4.400
-21UO	LSE61E-3	16	61	0.090	1.340	1.425	4.400
-22UO	LSE2E-4	14	2	0.045	0.415	0.470	2.770
-23UO	LSE3E-4	14	3	0.045	0.435	0.490	2.770
-24UO	LSE4E-4	14	4	0.045	0.470	0.525	2.770
-25UO	LSE7E-4	14	7	0.045	0.550	0.605	2.770
-26UO	LSE19E-6	12	19	0.075	0.975	1.055	1.740
-27UO	LSE2E-9	10	2	0.045	0.520	0.575	1.090
-28UO	LSE3E-9	10	3	0.045	0.545	0.600	1.090
-29UO	LSE4E-9	10	4	0.045	0.595	0.660	1.090
-30UO	LSE2E-14	9	2	0.045	0.595	0.660	0.868
-31UO	LSE3E-14	9	3	0.050	0.635	0.700	0.868
-32UO	LSE4E-14	9	4	0.055	0.705	0.770	0.868
-33UO	LSE19E-14	9	19	0.090	1.390	1.475	0.868
-34UO	LSE2E2-23	7	2	0.050	0.680	0.740	0.560
-35UO	LSE3E-23	7	3	0.055	0.720	0.795	0.560

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TABLE I. Details (non-watertight) – Continued.

Military part no. M24643/81	Type and size	Conductor size (AWG/ MCM)	Number of conductors	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)
					min.	max.	
-36UO	LSE4E-23	7	4	0.060	0.800	0.875	0.560
-37UO	LSE19E-23	7	19	0.090	1.585	1.690	0.560
-38UO	LSE4E-40	4	4	0.070	1.050	1.150	0.271
-39UO	LSE19E-40	4	19	0.090	2.020	2.125	0.271
-40UO	LSE2E-50	3	2	0.070	0.920	0.990	0.215
-41UO	LSE3E-50	3	3	0.075	0.980	1.060	0.215
-42UO	LSE19E-60	2	19	0.090	2.420	2.525	0.170
-43UO	LSE2E-75	1	2	0.085	1.135	1.220	0.136
-44UO	LSE3E-75	1	3	0.090	1.215	1.300	0.136
-45UO	LSE2E-100	0	2	0.090	1.220	1.305	0.108
-46UO	LSE3E-100	0	3	0.090	1.305	1.390	0.108
-47UO	LSE2E-150	000	2	0.090	1.345	1.430	0.068
-48UO	LSE3E-150	000	3	0.090	1.440	1.525	0.068
-49UO	LSE1E-300	300	1	0.090	0.970	1.050	0.038
-50UO	LSE1E-400	400 (Navy STD)	1	0.090	1.175	1.250	0.028
-51UO	LSE2E-400	400 (Navy STD)	2	0.090	2.020	2.220	0.029
-52UO	LSE3E-400	400 (Navy STD)	3	0.090	2.150	2.350	0.029

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TABLE II. Details (watertight).

Military part no. M24643/81	Type and size	Conductor size (AWG/ MCM)	Number of conductors	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)
					min.	max.	
-53UO	LSE7EW-2	18	7	0.045	0.510	0.560	7.070
-54UO	LSE10EW-2	18	10	0.045	0.620	0.685	7.070
-55UO	LSE14EW-2	18	14	0.050	0.675	0.740	7.070
-56UO	LSE19EW-2	18	19	0.055	0.745	0.820	7.070
-57UO	LSE24EW-2	18	24	0.065	0.880	0.955	7.070
-58UO	LSE30EW-2	18	30	0.070	0.930	1.005	7.070
-59UO	LSE37EW-2	18	37	0.075	1.005	1.090	7.070
-60UO	LSE44EW-2	18	44	0.085	1.135	1.220	7.070
-61UO	LSE2EW-3	16	2	0.045	0.420	0.470	4.400
-62UO	LSE3EW-3	16	3	0.045	0.435	0.490	4.400
-63UO	LSE4EW-3	16	4	0.045	0.465	0.520	4.400
-64UO	LSE7EW-3	16	7	0.045	0.540	0.590	4.400
-65UO	LSE10EW-3	16	10	0.050	0.670	0.735	4.400
-66UO	LSE14EW-3	16	14	0.055	0.725	0.795	4.400
-67UO	LSE19EW-3	16	19	0.060	0.805	0.880	4.400
-68UO	LSE24EW-3	16	24	0.070	0.950	1.025	4.400
-69UO	LSE30EW-3	16	30	0.075	1.005	1.090	4.400
-70UO	LSE33EW-3	16	33	0.080	1.045	1.130	4.400
-71UO	LSE37EW-3	16	37	0.085	1.090	1.175	4.400
-72UO	LSE44EW-3	16	44	0.090	1.230	1.315	4.400
-73UO	LSE61EW-3	16	61	0.090	1.375	1.460	4.400
-74UO	LSE2EW-4	14	2	0.045	0.445	0.500	2.770
-75UO	LSE3EW-4	14	3	0.045	0.465	0.520	2.770
-76UO	LSE4EW-4	14	4	0.045	0.500	0.555	2.770
-77UO	LSE7EW-4	14	7	0.045	0.580	0.640	2.770
-78UO	LSE19EW-6	12	19	0.075	1.010	1.095	1.740
-79UO	LSE2EW-9	10	2	0.045	0.550	0.505	1.090
-80UO	LSE3EW-9	10	3	0.045	0.575	0.635	1.090
-81UO	LSE4EW-9	10	4	0.050	0.630	0.695	1.090
-82UO	LSE2EW-14	9	2	0.050	0.630	0.695	0.868
-83UO	LSE3EW-14	9	3	0.050	0.670	0.735	0.868
-84UO	LSE4EW-14	9	4	0.055	0.735	0.810	0.868
-85UO	LSE19EW-14	9	19	0.090	1.425	1.510	0.868
-86UO	LSE2EW-23	7	2	0.055	0.715	0.715	0.560
-87UO	LSE3EW-23	7	3	0.055	0.755	0.825	0.560

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TABLE II. Details (watertight) – Continued.

Military part no. M24643/81	Type and size	Conductor size (AWG/ MCM)	Number of conductors	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)
					min.	max.	
-88UO	LSE4EW-23	7	4	0.065	0.835	0.910	0.560
-89UO	LSE19EW-23	7	19	0.090	1.625	1.725	0.560
-90UO	LSE19EW-40	4	19	0.090	2.055	2.160	0.271
-91UO	LSE2EW-50	3	2	0.065	0.955	1.030	0.215
-92UO	LSE3EW-50	3	3	0.075	1.105	1.190	0.215
-93UO	LSE19EW-60	2	19	0.090	2.455	2.560	0.170
-94UO	LSE2EW-75	1	2	0.075	1.170	1.255	0.136
-95UO	LSE3EW-75	1	3	0.090	1.255	1.335	0.136
-96UO	LSE2EW-100	0	2	0.090	1.255	1.340	0.108
-97UO	LSE3EW-100	0	3	0.090	1.345	1.425	0.108
-98UO	LSE3EW-150	000	3	0.090	1.440	1.525	0.068
-99UO	LSE1EW-300	300	1	0.090	1.005	1.090	0.038

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I or table II. No failure.
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	5,000
Conductor to overall shield	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	2,000
Conductor to overall shield	1,000
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160

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Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation	
Tensile strength (lb/in^2, min.)	75
Elongation (percent, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability (applicable to watertight constructions only)	No failure
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	5.0
Electromagnetic pulse (EMP) response (dB, min.)	80
Group D:	
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (86±3.6 °F [30±2 °C])	No damage
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	

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Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (overall cable jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure
Long-term jacket immersion	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	500

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-002)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

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